1) Diff between crude drug and pure drug ?
// // Non conventional and conventional

2) ~~pdf~~ give reason why is pravastatin produced through microbial transformation ?
gl why is compactin (mevastatin) used as precusor for pravastatin synthesis ?

compactin (mevastatin) —biotransformation by streptomyces carbophilus→ pravastatin

by hydroxylation at 6β position

(+) Advantges

pravastatin less that compactin in myopathy

(3) classification according to potency

Nux vomica contain strychnine which make tonic and clonic convulsion and Brucine

(4) why we use extract of plant ?

because Active constituent present in lower concentration and contain variety of constituent (active and inactive like cellulose, starch, sugar) so we use extract to control factors

Belladonna in winter different in summer

active constituent amount

5) give reason we use Rhubarb instead of senna?

because Rhubarb contain Anthraquinone which is laxative and tannin which is astringent so there is antagonism on digestive system on otherhand, senna is highly irritant laxative

6) is pyrrolizidine alkaloid is hepatotoxic ?

like senecionine in senecio

pyrrolizidine alkaloid is metabolized in liver by cytochrome P450 to dihydropyrolyzidine alkaloid which make alkylation for DNA causing carcinogenicity, hepatotoxicity and mutagenicity

7) some standarized herbal supp contain

1) Ginko (cerebral vasodilator) diterbene lactone ↓ cognitive function

2) St John wort hyperforin, antidepressent

standarized by Naphthodianthrone (hypericin)

standarization (standard content)

give reason

* كل dietary food should labelled by (dietary supplement) FDA قالتكدا

* SE or toxic reaction to herbal medicines ?

1) Mislabeling : proper identification with labeling.

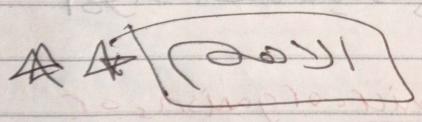
2) addition of unnatural synthetic substances ** chinese herbal remedies (Tumeric) بعض used for arthritis contain synthetic anti inflammatory such indomethacin , phenylbutazone
sideeffect on GIT and kidney SE : Thrombocytopenia
sideeffect toxic المواد الصناعية في

3) Natural toxic contaminants : microorganisms or their metabolity like aflatoxin → carcinogenic for liver

4) Natural toxic ingredient : Pyrrolizidine alkaloid hepatotoxic

contamination detection ; ---- سؤال استدم ايه في

1. ICP-MS (Inductively coupled plasma mass spectrometry)
   for :- heavy metals (lead, mercury, cadmium) in ginseng , tumeric
2. HPLC for fungal toxin (aflatoxin) liquid chrom
3. chromatographic (HPLC , LC-MS)
   for (Tumeric contained ibuprofen, steroid, Indomethacin , NSAID (synthetic drug) في النبات اي دواء صناعي



* ginseng contain TriTerpenoidal saponins (ginsenosides) excessive dose ← agitation , insomnia , raised blood pressure

question :- why ginseng not used with antidiabetic drug because ginsenosides decrease blood glucose so risk of hypoglycemia

give reason

liquorice Not with corticosteroid?
because it contains Triterpenoid sapponin (Glycyrrhizin) causes Na and water retention cause edema and increase BP.

liquorice Not with digitalis? ****
because Glycyrrhizin caus Na, water retention and hypokalemia, hypokalemia causes digitalis toxicity and cause arrthymia

liquorice Not with hypertensive drug?
نعني ادوية الكورتيزون

parsley contain apiole which related to safrole which is hepatocarinogen

KavaKava :- hepatotoxic in high dose

(apiole) parsley → hepatotoxic ليس
KavaKava / pyroliZidine (senecio)

ephedra :- contain ephedrine sympathomimetic cause hypertention, stroke, death

chamomile → sesquiterpene lactones as matricin → allergy

*** parsley contains furanocoumarin as psoralens which cause phototoxic reaction

psoralen يستخرج vitiligo علاج → phototoxicity يسبب

star anise ~~***~~ (antispasmodic, carminative)

chinese star anise is safe ✓
Japanese contain sikimitoxin which is neurotoxin (GABA antagonist cause seizure, rapid eye movement, agitation, nausea, vomiting.

يشبهان morphologically No diff

Herb DE